

Milestone 3 – Geographical Analysis of US Disaster Declarations

This milestone analyzes the geographical distribution of natural disaster declarations across the United States.

The objective is to identify disaster hotspots and visualize disaster patterns across different states.

STEP1. Import libraries and load the cleaned dataset

```
In [5]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

df = pd.read_csv("usnd_cleaned.csv")
df.head()
```

Out[5]: fema_declaration_string disaster_number state declaration_type declaration_date fy_d

0	DR-1-GA	1	GA	DR	1953-05-02 00:00:00+00:00
1	DR-2-TX	2	TX	DR	1953-05-15 00:00:00+00:00
2	DR-3-LA	3	LA	DR	1953-05-29 00:00:00+00:00
3	DR-4-MI	4	MI	DR	1953-06-02 00:00:00+00:00
4	DR-5-MT	5	MT	DR	1953-06-06 00:00:00+00:00

5 rows × 25 columns



This loads the clean dataset from Milestone-1 which contains the processed disaster data

Here we loaded the cleaned FEMA disaster dataset

```
In [6]: df.shape
```

```
Out[6]: (68485, 25)
```

```
In [7]: df.isnull().sum()
```

```
Out[7]: fema_declaration_string      0
disaster_number                    0
state                              0
declaration_type                    0
declaration_date                    0
fy_declared                        0
incidentType                        0
declaration_title                    0
ih_program_declared                 0
ia_program_declared                 0
pa_program_declared                 0
hm_program_declared                 0
incident_begin_date                 0
tribal_request                      0
fips                                0
place_code                          0
designated_area                      0
declaration_request_number          0
incident_id                         0
region                              0
last_refresh                        0
hash                                0
id                                  0
year                                0
month                               0
dtype: int64
```

The target variable in Milestone-3 is: disaster_count. It represents the number of disaster declarations in each state. This variable is derived using aggregation.

Renamed the required columns

```
In [8]: df.rename(columns={'incident_type':'incidentType'}, inplace=True)

df['declaration_date'] = pd.to_datetime(df['declaration_date'])

df['year'] = df['declaration_date'].dt.year
df['month'] = df['declaration_date'].dt.month
```

```
In [9]: df.to_csv("usnd_cleaned.csv", index=False)
df
```

Out[9]:

	fema_declaration_string	disaster_number	state	declaration_type	declaration_date
0	DR-1-GA	1	GA	DR	1953-05-02 00:00:00+00:00
1	DR-2-TX	2	TX	DR	1953-05-15 00:00:00+00:00
2	DR-3-LA	3	LA	DR	1953-05-29 00:00:00+00:00
3	DR-4-MI	4	MI	DR	1953-06-02 00:00:00+00:00
4	DR-5-MT	5	MT	DR	1953-06-06 00:00:00+00:00
...	...	...	...	...	...
68480	DR-4892-WI	4892	WI	DR	2025-09-11 00:00:00+00:00
68481	DR-4892-WI	4892	WI	DR	2025-09-11 00:00:00+00:00
68482	DR-4890-SD	4890	SD	DR	2025-09-11 00:00:00+00:00
68483	FM-5613-HI	5613	HI	FM	2025-09-23 00:00:00+00:00
68484	FM-5614-WA	5614	WA	FM	2025-09-26 00:00:00+00:00

68485 rows × 25 columns



STEP2-Verified Required columns.

This selects specific columns from the dataset.

```
In [10]: df[['state', 'incidentType', 'year', 'month']].head()
```

```
Out[10]:
```

	state	incidentType	year	month
0	GA	Tornado	1953	5
1	TX	Tornado	1953	5
2	LA	Flood	1953	5
3	MI	Tornado	1953	6
4	MT	Flood	1953	6

STEP-3 State-level Aggregation

Groups the dataset based on state.

```
In [11]: state_counts = df.groupby('state').size().reset_index(name='disaster_count')
state_counts.sort_values(by='disaster_count', ascending=False).head()
```

```
Out[11]:
```

	state	disaster_count
50	TX	5386
20	KY	3235
28	MO	2829
10	FL	2791
12	GA	2653

Inference: This step helps identify which states experience the highest number of disasters

STEP-4 State + Incident Type Aggregation

Identify which disaster types occur more in each state.

```
In [12]: state_incident = df.groupby(['state', 'incidentType']).size().reset_index(name='count')
state_incident.head()
```

Out[12]:

	state	incidentType	count
0	AK	Biological	121
1	AK	Coastal Storm	2
2	AK	Earthquake	13
3	AK	Fire	35
4	AK	Flood	61

STEP-5 Sort Disaster Counts

Sorts the states based on disaster count

```
In [13]: state_counts.sort_values(by='disaster_count', ascending=False).head()
```

Out[13]:

	state	disaster_count
50	TX	5386
20	KY	3235
28	MO	2829
10	FL	2791
12	GA	2653

Inference: Southern and coastal states often have more disasters due to hurricanes, floods and storms.

STEp-6 Top10 Disaster States (Bar chat)- Disaster Distribution.

```
In [37]: # Get top 10 states
top_states = df['state'].value_counts().head(10).index

filtered_df = df[df['state'].isin(top_states)]

state_incident = filtered_df.groupby(['state', 'incidentType']).size().reset_index(n

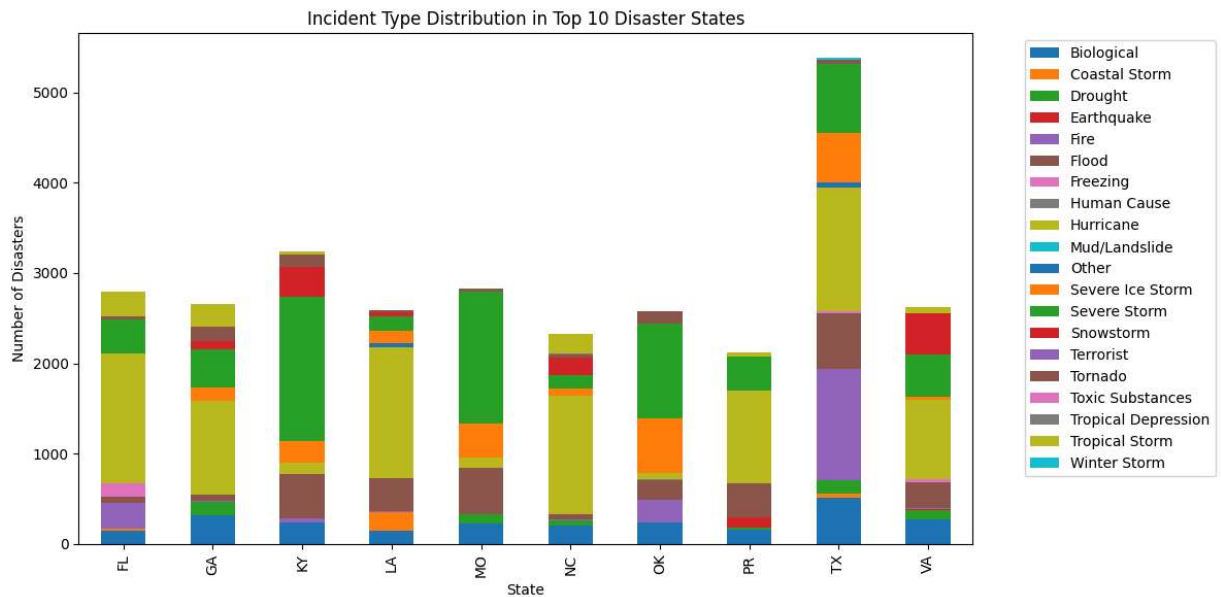
pivot_data = state_incident.pivot(index='state', columns='incidentType', values='co

pivot_data.plot(kind='bar', stacked=True, figsize=(12,6))

plt.title("Incident Type Distribution in Top 10 Disaster States")
plt.xlabel("State")
plt.ylabel("Number of Disasters")
```

```
plt.xticks(rotation=90)
plt.legend(bbox_to_anchor=(1.05,1), loc='upper left')

plt.tight_layout()
plt.show()
```



The stacked bar chart shows the distribution of different disaster types across the top disaster-prone states. Certain states such as Texas and California experience multiple disaster types including storms, floods, and fires. Different geographic and climatic conditions in these states make them vulnerable to various natural disasters

Inference: Different states experience different types of disasters.

STEP-7 Choropleth Map (Main Visualization)

```
In [14]: fig = px.choropleth(
    state_counts,
    locations='state',
    locationmode="USA-states",
    color='disaster_count',
    scope="usa",
    color_continuous_scale="Oranges",
    title="US Disaster Declarations by State"
)

fig.show()
```

Explanation: This map colors states according to disaster frequency. Darker color → more disasters. Insight example: Observed pattern : Southern states such as Texas and Florida show higher disaster counts. Reason: These regions experience hurricanes, floods and tropical storms.

STEP-8 Disaster Type Hotspot Map

```
In [15]: hurricane_data = df[df['incidentType']=="Hurricane"]

hurricane_counts = hurricane_data.groupby('state').size().reset_index(name='count')

fig = px.choropleth(
    hurricane_counts,
    locations='state',
    locationmode="USA-states",
    color='count',
    scope="usa",
    title="Hurricane Disaster Hotspots"
)

fig.show()
```

Inference: This visualization helps identify geographical disaster hotspots.

Milestone-3 analysis shows that disaster declarations are not evenly distributed across the United States.

Certain states such as Texas, California, and Florida experience significantly more disasters due to their geographical and climatic conditions.

The visualizations help identify disaster hotspots and dominant disaster types, which can assist in disaster preparedness and policy planning